

Thabang L. Mashinini-Sekgoto

LEAD DATA SCIENTIST (ACTING HEAD OF DATA SCIENCE) · AI & ANALYTICS LEADER

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Summary

AI and Data Science Leader with 10+ years of experience operationalising enterprise machine learning across banking, insurance, and telecommunications. Currently Acting Head of Data Science at ABSA Insurance, leading a hyperpersonalisation intelligence platform processing **2M+** daily trips across **26,000+** telematics policyholders, and geospatial climate-risk models spanning **230,000+** insured properties. Previously led Vodacom's Smart Generator Optimisation platform, directly contributing to **≈R1 billion** in annual operational savings across **15,000+** sites. NeurIPS-published climate ML researcher (IBM Research) whose PhD research at Wits focuses on physics-informed self-supervised learning for flood extent mapping. Founder of **Ubunye Engine**, an open-source Spark-native AI/ETL framework published on PyPI. Core expertise in Databricks, distributed streaming, LLM architectures, and enterprise AI governance in regulated environments.

Education

University of the Witwatersrand

Johannesburg, South Africa

PHD IN COMPUTER SCIENCE (IN PROGRESS)

2024 – Present

- Research: Physics-informed self-supervised learning for SAR-based flood extent mapping, with applications to insurance risk and data-scarce regions.
- Intersection of remote sensing, self-supervised learning, and computational hydrology.

University of the Witwatersrand

Johannesburg, South Africa

MSC IN COMPUTER SCIENCE (DISTINCTION)

2018 – 2019

- Thesis: Learning Level Set Method by Echo State Network for Image Segmentation.
- Proposed a novel spatiotemporal deep-learning formulation of variational level set segmentation, benchmarking five recurrent and convolutional architectures across four datasets.

University of the Witwatersrand

Johannesburg, South Africa

BSC HONOURS IN COMPUTER SCIENCE

2017

- Project: Wildfire Estimation Using Kernel Density Estimators.

University of the Witwatersrand

Johannesburg, South Africa

BSC IN COMPUTATIONAL & APPLIED MATHEMATICS AND ASTRONOMY

2014 – 2016

- Focused on mathematical modelling, astrophysics, simulation, and numerical methods.

Experience

ABSA Insurance

Johannesburg, South Africa

LEAD DATA SCIENTIST (ACTING HEAD OF DATA SCIENCE)

Mar. 2024 – Present

- Provide executive functional leadership for the Insurance Data Science capability—defining analytical priorities, technical direction, MLOps standards, and AI governance across underwriting, retention, fraud, telematics, and climate risk.
- Lead a multidisciplinary team of senior data scientists and data engineers while orchestrating cross-functional delivery across business, technology, risk, architecture, and executive stakeholders.
- Formulated and executed the enterprise Data Engineering and MLOps strategy on Databricks, building modern data pipelines, automated orchestration, and governance frameworks that accelerated model time-to-production.
- Champion ABSA Insurance's hyperpersonalisation vision, architecting an intelligence platform that processes **2M+** daily telematics signals to drive retention, next-best-action, and Customer Lifetime Value (CLV).
- Modernised the core telematics platform using the Ubunye Engine, reducing data pipeline processing latency from **2 months to under 24 hours** at scale.
- Developed high-resolution geospatial flood and natural-catastrophe risk models using XGBoost and JBA ground-truth data across **230,000+** insured properties to optimize underwriting.
- Institutionalised enterprise AI governance and CI/CD pipelines using Databricks Asset Bundles, MLflow, and Unity Catalog across a production portfolio of all our analytical models.

Vodacom

Johannesburg, South Africa

SENIOR DATA SCIENTIST

Nov. 2021 – Mar. 2024

- Led a high-performing team of **10** (9 Data Scientists, 1 Senior ML Engineer) engineering real-time analytics and optimisation platforms for national telecommunications infrastructure.
- Architected the Smart Generator Optimisation platform across **15,000+** national infrastructure sites, directly contributing to **≈R1 billion** in annual operational cost savings.
- Built high-throughput streaming pipelines processing **25M+** daily telemetry events from **15,000+** sites and mobile generators via Kafka, PySpark, and Kubernetes.
- Designed a custom PyFlink stream-processing framework to enrich real-time Kafka topics with IoT telemetry, operational alarms, and static reference data on the fly.
- Institutionalised enterprise AI and analytics engineering standards across the team, streamlining model development cycles and system reliability.
- Honoured with the Vodacom Star Award in recognition of outstanding engineering

IBM Research

Johannesburg, South Africa

MACHINE LEARNING RESEARCH SCIENTIST

Apr. 2020 – Nov. 2021

- Developed machine learning and geospatial analytics solutions for environmental intelligence and climate-risk applications.
- Built predictive analytics systems using TensorFlow and distributed data platforms.
- Contributed to the Gauteng COVID-19 risk-index and prediction dashboard (IBM Research Africa, Wits University, and GCRO), used by the Gauteng Provincial Department of Health to identify infection hotspots and forecast hospital bed and ventilator demand for a province of **15M+** residents; deployed publicly via IBM Cloud.
- Deployed climate-forecasting models into the IBM PAIRS Geoscope platform, an enterprise geospatial-temporal system handling petabyte-scale data for clients globally.
- Co-authored peer-reviewed research and collaborated with international research and engineering teams across large-scale AI initiatives.

Business Intelligence Services – University of the Witwatersrand

Johannesburg, South Africa

DATA SCIENTIST

Jun. 2018 – Apr. 2020

- Developed recommendation and analytics systems supporting institutional planning and student success initiatives.
- Built reporting and analytics workflows using Python, SQL, and Power BI.
- Developed a clustering-based recommendation engine serving the Faculty of Humanities (**10,000+** students), generating over **R2 million** annually in government subsidy impact.
- Facilitated **7** analytics and machine learning workshops, training **76+** technical and non-technical staff and students.

Council for Scientific and Industrial Research (CSIR)

Pretoria, South Africa

DATA SCIENTIST AND SOFTWARE ENGINEER

Nov. 2017 – Jan. 2018

- Developed predictive analytics and operational intelligence systems supporting municipalities and public-sector decision-making initiatives.
- Built Django-based decision-support systems serving **17** municipalities — including the City of Cape Town and **16** across Gauteng — enabling operational visibility and real-time access to analytics for public-sector stakeholders.
- Applied machine learning and data engineering techniques to identify operational bottlenecks and improve service delivery planning processes.
- Collaborated within multidisciplinary teams combining software engineering, analytics, and public-sector innovation initiatives.
- Recognised for innovation in predictive modelling and enterprise solutions by  [Mail & Guardian](#) ,  [CSIR](#) , and  [Department of Science and Technology \(DST\)](#) .

ZAR X

Johannesburg, South Africa

SYSTEM ANALYST & SUPPORT

Mar. 2017 – Nov. 2017

- Provided systems analysis and operational support for a newly launched stock exchange serving **13,000+** clients, supporting trading platform stability during go-live and early operations.

Skills

Leadership	AI Strategy, Technical Leadership, Stakeholder Management, Team Mentorship, Agile Delivery
Programming	Python, SQL, R, JavaScript, C/C++, Shell Scripting
AI & Machine Learning	Machine Learning, Predictive Analytics, LLMs & RAG, Agentic AI Systems, NLP, Computer Vision
Data Platforms	Databricks, Apache Spark, Kafka, Flink, Hadoop, ETL/ELT Pipelines
Analytics	Power BI, Dashboarding, Data Visualisation, Business Intelligence, Statistical Analysis
MLOps & Engineering	MLflow, Docker, Kubernetes, CI/CD, GitHub Actions, Real-time ML Systems
Generative AI	LLM Application Development (Claude, Gemini, OpenAI APIs), RAG Architectures, Multi-Model Orchestration, Custom AI Agents & Gateways, AI-Assisted Engineering (Claude Code)
Cloud & Infrastructure	AWS, Kubernetes, Linux, Distributed Systems, Serverless Deployment (Vercel), Cloud-native AI Platforms

Publications

Long-Range Seasonal Forecasting of 2m-Temperature with Machine Learning

NEURIPS 2020

2020

- Published research focused on probabilistic forecasting and machine learning methods for climate intelligence systems.

ML-based Probabilistic Prediction of 2m Temperature and Total Precipitation

EGU 2022

2022

- Research focused on machine learning approaches for probabilistic climate forecasting.

Mine Worker Threshold Shift Estimation via Optimization Algorithms for Deep Recurrent Neural Networks

IFAC MMM 2019

2019

- Research focused on optimisation and recurrent neural networks in mining applications.

Selected Projects

AI Knowledge Platform & Virtual AI Mentor

AI EDUCATOR

2024 – Present

- Established a public AI leadership and technical education platform focused on accelerating practical AI adoption, technical mentorship, and industry-wide AI literacy.
- Co-host **FabAcademic Unfiltered** ([🔗 YouTube](#)) alongside *Prof. Mamokgethi Phakeng* ([🔗 profile](#)), delivering strategic AI insights to a global audience of **300,000+** across platforms.
- Publish technical leadership articles generating **12,000+** monthly visits, distilling complex topics in distributed systems, MLOps, AI governance, and enterprise machine learning architecture.
- Architected and deployed **Thabang AI** ([🔗 Platform](#)), a RAG-grounded serverless assistant utilizing a multi-model architecture (Claude, Gemini, OpenAI), custom AI gateway, and agentic workflows for technical career guidance.

Ubunye AI Ecosystems

FOUNDER & ARCHITECT

2024 – Present

- Designed and architected the **Ubunye Engine** an open-source, config-driven distributed AI and ETL framework designed to standardize fragmented machine learning and data engineering workflows.
- Built a modular, multi-backend engine supporting **PySpark, Dask, and Pandas**, enabling scalable feature engineering, reproducible ML pipelines, and production-grade analytics.
- Published to PyPI and integrated into enterprise data-engineering workstreams, reducing pipeline prototyping-to-deployment time from **≈2 weeks to under 8 hours**.
- Authored public documentation, architectural guides, and interactive Databricks notebooks to accelerate community and enterprise adoption.
- [🔗 Documentation & Open Source Repository](#)

Tfilterspy

CREATOR & PRINCIPAL DEVELOPER

2023 – Present

- Designed and open-sourced **tfilterspy**, a distributed Bayesian filtering and probabilistic state-estimation framework built for noisy real-world telemetry, IoT, and time-series systems.
- Implemented parallelised Kalman and Particle filtering architectures on **Dask**, enabling scalable probabilistic state-estimation across high-throughput streaming sensor data.
- Engineered state-estimation algorithms specifically designed to tackle signal noise, missing telemetry, and sensor degradation in production insurance and telecommunications applications.
- [🔗 Documentation & Open Source Repository](#)

Smart Generator Optimisation Platform

SENIOR DATA SCIENTIST

2021 – 2024

- Co-founded digital transformation initiatives aimed at bridging the digital divide for township SMEs and local entrepreneurs by expanding their online market presence.
- Designed and deployed lightweight, highly accessible React-based web platforms, successfully onboarding **17** local businesses into the formal digital economy.
- Built scalable, low-cost digital infrastructure tailored to low-bandwidth environments, driving practical business enablement and removing friction for digital adoption.
- [🔗 Project Website](#)

Kasi'Lam Digital Platforms

Co-FOUNDER

2022 – Present

- Co-founded digital transformation initiatives aimed at bridging the digital divide for township SMEs and local entrepreneurs by expanding their online market presence.
- Designed and deployed lightweight, highly accessible React-based web platforms, successfully onboarding **17** local businesses into the formal digital economy.
- Built scalable, low-cost digital infrastructure tailored to low-bandwidth environments, driving practical business enablement and removing friction for digital adoption.
-  [Project Website](#)

Honors & Awards

2022	Vodacom Star Award , Recognised for innovation and engineering excellence	<i>Vodacom</i>
2019	MSc with Distinction , Outstanding academic and research performance	<i>University of the Witwatersrand</i>
2014 – 2017	Co-Founder – Wits Astronomy Club , STEM outreach and astronomy public engagement initiatives	<i>Wits University</i>

References

Dr.	Etienne Vos (IBM Research Manager) ,  Etienne.vos@ibm.com	<i>IBM Research</i>
Dr.	Jacobus Du Toit (Machine Learning Lead) ,  jacoDutoit@vodacom.co.za	<i>Vodacom</i>
Dr.	Akram Zaytar (Senior Research Scientist) ,  akramzaytar@microsoft.com	<i>IBM/Microsoft Research</i>
Mrs.	Harshila Dulabh (Team Lead) ,  Harshila.Dulabh@wits.ac.za	<i>Wits University (BIS)</i>